Expanded Mathematical Learning Report: American Option Risk Surfaces and Modular Neural Diagnostics

Free-Boundary-Aware Risk Surfaces for American Options

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Abstract

This report is the learning companion to the concise final report. It explains the mathematical formulation, finite-difference matrices, CN/PSOR calculation, boundary and Greek diagnostics, dataset governance, and modular neural diagnostics for a second-year AI/ML undergraduate. It introduces no new experiments or numerical claims.

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How to Use This Learning Version

Read Stages 1–2 for the mathematical and numerical derivation. Stages 3–4 explain how secondary quantities inherit grid limitations. Stages 5–6 explain how solver outputs become governed data. Stages 7–11 explain why price, boundary, and Delta are modeled and audited separately.

1 Stage 1: Mathematical Foundations

1.1 Purpose

This stage defines the financial problem and the notation used throughout the project. The objective is not to introduce a new option-pricing theory. It is to explain why early exercise changes a familiar Black–Scholes equation into an inequality problem with an unknown boundary.

1.2 Method

Let $S > 0$ denote the current asset price, $K > 0$ the strike, T the maturity date, $t \in [0, T]$ calendar time, r the continuously compounded risk-free rate, q the continuous dividend yield, and $\sigma > 0$ the volatility. The project studies American puts and American calls. Their immediate-exercise payoffs are

$$\Phi_{\text{put}}(S) = \max(K - S, 0), \qquad \Phi_{\text{call}}(S) = \max(S - K, 0).$$

The American value $V(S, t)$ is the best discounted expected payoff available when the holder may choose an exercise time θ between t and T:

$$V(S,t) = \sup_{\theta \in [t,T]} \mathbb{E}^{\mathbb{Q}}\left[e^{-r(\theta - t)} \Phi(S_\theta) \,\middle|\, S_t = S \right].$$

The expectation is under the risk-neutral measure $\mathbb{Q}$.

> **Reader intuition.** A European holder has one allowed exercise date. An American holder repeatedly compares two actions: exercise now for $\Phi(S)$, or retain the contract. The option value must therefore be at least the immediate payoff.

1.3 Mathematical Formulation

1.3.1 Time to maturity

The numerical solver uses

$$\tau = T - t, \qquad U(S, \tau) = V(S, T - \tau).$$

Thus $\tau = 0$ is maturity and $\tau = T$ is the initial valuation date. This change converts a terminal-value problem in t into an initial-value problem in τ:

$$U(S, 0) = \Phi(S).$$

> **Common confusion.** Increasing t moves forward in calendar time, but increasing τ moves away from maturity. The code begins with the payoff at $\tau = 0$ and advances through $\tau_1, \tau_2, \ldots, \tau_N = T$.

1.3.2 Risk-neutral dynamics and the Black–Scholes operator

Under the model used in the project,

$$dS_t = (r - q)S_t\, dt + \sigma S_t\, dW_t.$$

In a region where exercise is not optimal, standard replication and risk-neutral valuation give

$$U_\tau = \mathcal{L}U,$$

where

$$\mathcal{L}U = \frac{1}{2}\sigma^2 S^2 U_{SS} + (r - q)SU_S - rU.$$

The three terms have separate meanings:

- $\frac{1}{2}\sigma^2 S^2 U_{SS}$ is the diffusion or curvature term;
- $(r - q)SU_S$ is the risk-neutral drift term;
- $-rU$ is the discount term.

1.3.3 From a PDE to a variational inequality

Immediate exercise is always available, so

$$U(S, \tau) \geq \Phi(S).$$

If $U(S, \tau) > \Phi(S)$, retaining the option is more valuable and the continuation PDE applies:

$$U_\tau - \mathcal{L}U = 0.$$

If $U(S, \tau) = \Phi(S)$, the holder exercises. At such a point the PDE equality need not hold, but delaying exercise cannot produce a locally better value. The combined problem is

$$U(S, \tau) \geq \Phi(S), \qquad U_\tau - \mathcal{L}U \geq 0, \qquad \big(U - \Phi\big)\big(U_\tau - \mathcal{L}U\big) = 0.$$

Reader intuition. The product condition says that at least one factor is zero. In continuation, $U - \Phi > 0$, so the PDE residual must be zero. In exercise, $U - \Phi = 0$, while the residual may be positive. This is why American pricing is an inequality or complementarity problem rather than only a linear PDE solve.

1.3.4 Exercise region, continuation region, and free boundary

Define

$$\mathcal{E} = \{(S, \tau) : U(S, \tau) = \Phi(S)\}, \qquad \mathcal{C} = \{(S, \tau) : U(S, \tau) > \Phi(S)\}.$$

The interface between $\mathcal{E}$ and $\mathcal{C}$ is the free boundary $B(\tau)$. It is "free" because its location is not prescribed as an input; it must be discovered together with the option value.

For a put, exercise generally occupies the low-spot side:

$$S \lesssim B_{\text{put}}(\tau) \quad \Longrightarrow \quad \text{exercise.}$$

For a dividend-paying call, exercise—when it occurs—generally occupies the high-spot side:

$$S \gtrsim B_{\text{call}}(\tau) \quad \Longrightarrow \quad \text{exercise.}$$

A no-dividend American call is retained as a control because, under the standard assumptions, early exercise is not optimal.

1.3.5 Continuation premium

The continuation premium is

$$C(S, \tau) = U(S, \tau) - \Phi(S).$$

The obstacle condition immediately gives

$$C(S, \tau) \geq 0.$$

Moreover,

$$C = 0 \quad \text{in the exercise region,} \qquad C > 0 \quad \text{in the continuation region.}$$

Reader intuition. C is the extra value of keeping the option alive instead of exercising immediately. It is therefore both a financial quantity and a useful numerical indicator of the exercise boundary.

Common confusion. Nonnegative continuation premium is only the payoff-obstacle requirement. It does not by itself guarantee monotonicity, convexity, a correct free boundary, or accurate Delta and Gamma.

1.4 Implementation Details

The solver stores spot along one array axis and time to maturity along the other. Reports use normalized quantities such as S/K, U/K, and B/K where useful, but the mathematical obstacle remains $U \geq \Phi$. Later neural features include

$$x = \big(\log(S/K), \tau/T, r, q, \sigma, T, \mathbf{1}_{\text{call}}\big).$$

1.5 Validation and Diagnostics

Before discretization, the notation is checked for the direction of time, the sign of $U_\tau - \mathcal{L}U$, put/call payoff orientation, and the units of $\Delta = U_S$ and $\Gamma = U_{SS}$. A sign error can yield stable-looking arrays while solving the wrong problem.

1.6 Results

Stage 1 produces a consistent mathematical contract. It introduces no new numerical results and makes no model-performance claim.

1.7 Failure Modes and Limitations

The formulation assumes the Black–Scholes diffusion with constant parameters. It does not include jumps, stochastic volatility, transaction costs, or market calibration. These omissions delimit the project rather than invalidate its within-model numerical study.

1.8 Conclusion and Next Step

The continuous problem is an obstacle-constrained PDE. Stage 2 converts its spatial operator into a tridiagonal matrix, applies Crank–Nicolson in time, and solves the resulting linear complementarity problem with PSOR.

2 Stage 2: From the PDE to CN/PSOR

2.1 Purpose

This stage explains the finite-difference matrix construction in enough detail to connect every mathematical coefficient with the arrays used by the solver. The target is a reader who understands derivatives and matrix multiplication but has not previously implemented an American-option LCP.

2.2 Method

2.2.1 Spot and time grids

Truncate the unbounded spot domain to $[0, S_{\max}]$. Choose M equal spot intervals and N equal time intervals:

$$S_i = i\Delta S, \qquad i = 0, \ldots, M, \qquad \Delta S = \frac{S_{\max}}{M},$$

$$\tau_n = n\Delta\tau, \qquad n = 0, \ldots, N, \qquad \Delta\tau = \frac{T}{N}.$$

Write U_i^n for the approximation to $U(S_i, \tau_n)$. At a fixed time, the interior unknown vector is

$$U^n = (U_1^n, U_2^n, \ldots, U_{M-1}^n)^\top.$$

Common confusion. The index i moves across spot and the index n moves across time to maturity. U_i^n is one scalar. U^n is the column vector containing all interior spot values at time level n.

Common confusion. U_0^n and U_M^n are not missing unknowns. They are values at the two domain boundaries $S = 0$ and $S = S_{\max}$, where an option-specific boundary condition supplies the value.

2.3 Mathematical Formulation

Step-by-step derivation: central differences

At an interior node, central differences give

$$U_S(S_i, \tau_n) \approx \frac{U_{i+1}^n - U_{i-1}^n}{2\Delta S},$$

$$U_{SS}(S_i, \tau_n) \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{(\Delta S)^2}.$$

Substitute them into

$$\mathcal{L}U = \frac{1}{2}\sigma^2 S^2 U_{SS} + (r - q)SU_S - rU.$$

At S_i,

$$(L_h U)_i = \frac{1}{2}\sigma^2 S_i^2 \frac{U_{i+1} - 2U_i + U_{i-1}}{(\Delta S)^2} + (r - q)S_i \frac{U_{i+1} - U_{i-1}}{2\Delta S} - rU_i.$$

Collect the coefficients of the three neighboring values:

$$(L_h U)_i = a_i U_{i-1} + b_i U_i + c_i U_{i+1},$$

where

$$a_i = \frac{\frac{1}{2}\sigma^2 S_i^2}{(\Delta S)^2} - \frac{(r-q)S_i}{2\Delta S},$$

$$b_i = -\frac{\sigma^2 S_i^2}{(\Delta S)^2} - r,$$

$$c_i = \frac{\frac{1}{2}\sigma^2 S_i^2}{(\Delta S)^2} + \frac{(r-q)S_i}{2\Delta S}.$$

2.3.1 Where the coefficients come from

Source	Contribution to a_i	Contribution to b_i	Contribution to c_i
Diffusion	$+\frac{1}{2}\sigma^2 S_i^2/(\Delta S)^2$	$-\sigma^2 S_i^2/(\Delta S)^2$	$+\frac{1}{2}\sigma^2 S_i^2/(\Delta S)^2$
Drift	$-(r-q)S_i/(2\Delta S)$	0	$+(r-q)S_i/(2\Delta S)$
Discount	0	$-r$	0

> **Reader intuition.** Diffusion spreads weight symmetrically to the left and right. Drift shifts the two off-diagonal weights in opposite directions. Discounting affects only the current node and therefore appears only in the diagonal coefficient.

Step-by-step derivation: a small matrix with $M = 4$

With nodes S_0, S_1, S_2, S_3, S_4, there are three interior unknowns:

$$U^n = (U_1^n, U_2^n, U_3^n)^\top.$$

The interior matrix is

$$L_h = \begin{bmatrix} b_1 & c_1 & 0 \\ a_2 & b_2 & c_2 \\ 0 & a_3 & b_3 \end{bmatrix}.$$

The full stencil equations are

$$(L_h U)_1 = a_1 U_0 + b_1 U_1 + c_1 U_2,$$
$$(L_h U)_2 = a_2 U_1 + b_2 U_2 + c_2 U_3,$$
$$(L_h U)_3 = a_3 U_2 + b_3 U_3 + c_3 U_4.$$

Matrix multiplication by the displayed 3×3 matrix contains only the interior terms. The omitted $a_1 U_0$ and $c_3 U_4$ are known boundary contributions.

> **Common confusion.** The first matrix row has no column for U_0, and the last row has no column for U_4, because the matrix maps the three-component interior vector to another three-component vector. Only the first stencil touches the lower boundary; only the last stencil touches the upper boundary.

Step-by-step derivation: Crank–Nicolson time stepping

Crank–Nicolson averages the spatial operator at the old and new time levels:

$$\frac{U^{n+1}-U^n}{\Delta\tau}=\frac{1}{2}L_hU^{n+1}+\frac{1}{2}L_hU^n.$$

Multiply by $\Delta\tau$:

$$U^{n+1}-U^n=\frac{\Delta\tau}{2}L_hU^{n+1}+\frac{\Delta\tau}{2}L_hU^n.$$

Move the new-time operator to the left:

$$U^{n+1}-\frac{\Delta\tau}{2}L_hU^{n+1}=U^n+\frac{\Delta\tau}{2}L_hU^n.$$

Define

$$A=I-\frac{\Delta\tau}{2}L_h,\qquad B=I+\frac{\Delta\tau}{2}L_h.$$

Then

$$AU^{n+1}=BU^n+g^{n,n+1},$$

where $g^{n,n+1}$ collects known boundary values.

> **Common confusion.** A multiplies U^{n+1} because its terms were gathered on the left. B multiplies U^n because its terms remain on the right. The signs are a consequence of rearrangement, not an arbitrary naming convention.

2.3.2 What one row means

Ignoring the boundary adjustment momentarily, an interior row i reads

$$\begin{aligned}-\frac{\Delta\tau}{2}a_iU_{i-1}^{n+1}+\left(1-\frac{\Delta\tau}{2}b_i\right)U_i^{n+1}-\frac{\Delta\tau}{2}c_iU_{i+1}^{n+1}\\=\frac{\Delta\tau}{2}a_iU_{i-1}^{n}+\left(1+\frac{\Delta\tau}{2}b_i\right)U_i^{n}+\frac{\Delta\tau}{2}c_iU_{i+1}^{n}.\end{aligned}$$

This is the scalar equation represented by row i of A and B.

Step-by-step derivation: why boundary values appear on the right-hand side

For the first interior row, the lower-boundary terms at the two time levels are

$$a_1U_0^{n+1},\qquad a_1U_0^n.$$

Moving the new-time boundary term to the right gives the combined contribution

$$\frac{\Delta\tau}{2}a_1\left(U_0^n+U_0^{n+1}\right).$$

Similarly, the last row receives

$$c_{M-1}U_M^{n+1}, \qquad c_{M-1}U_M^n,$$

which combine as

$$\frac{\Delta\tau}{2}c_{M-1}\left(U_M^n + U_M^{n+1}\right).$$

Hence

$$g^{n,n+1} = \frac{\Delta\tau}{2}\begin{bmatrix} a_1(U_0^n + U_0^{n+1}) \\ 0 \\ \vdots \\ 0 \\ c_{M-1}(U_M^n + U_M^{n+1}) \end{bmatrix}.$$

Implementation connection. This matches `src/solvers/cn_psor.py`: the first right-hand-side entry is increased by $\frac{1}{2}\Delta\tau\, a_1(U_0^n + U_0^{n+1})$, and the last by $\frac{1}{2}\Delta\tau\, c_{M-1}(U_M^n + U_M^{n+1})$.

2.3.3 Boundary conditions used in the project

For an American put,

$$U(0,\tau) = K, \qquad U(S_{\max},\tau) = 0.$$

At zero spot, immediate exercise returns K. At a sufficiently large spot, the put value approaches zero.

For a dividend-paying American call,

$$U(0,\tau) = 0,$$

$$U(S_{\max},\tau) = \max\left(S_{\max} - K,\, S_{\max}e^{-q\tau} - Ke^{-r\tau},\, 0\right).$$

The upper boundary retains both immediate-exercise value and the discounted European far-field expression.

For European validation with projection disabled, the put uses

$$U(0,\tau) = Ke^{-r\tau}, \qquad U(S_{\max},\tau) = 0,$$

and the call uses

$$U(0,\tau) = 0, \qquad U(S_{\max},\tau) = S_{\max}e^{-q\tau} - Ke^{-r\tau}.$$

Step-by-step derivation: the American linear complementarity problem

Let

$$d^n = BU^n + g^{n,n+1}.$$

A European step solves the equality $AU^{n+1} = d^n$. For an American option, that equality can produce a value below the payoff. The discrete obstacle problem is instead

$$AU^{n+1} \geq d^n, \qquad U^{n+1} \geq \Phi,$$

$$\left(U^{n+1} - \Phi\right)^{\top} \left(AU^{n+1} - d^n\right) = 0.$$

The first inequality requires a nonnegative discrete PDE residual. The second enforces the payoff obstacle. The final equality imposes complementarity: each node is either continuing with zero PDE residual or exercising with zero premium.

> **Common confusion.** An American option is not priced by solving the European system and then reporting whichever nodes happen to lie above the payoff. The obstacle is part of the system itself, so the solution method must enforce it during the solve.

Step-by-step derivation: projected successive over-relaxation

At one time step, write $Ax = d$ for the interior system and let Φ_i be the obstacle at node i. Starting from an old iterate $x^{(k)}$, a Gauss–Seidel candidate is

$$y_i^{(k+1)} = \frac{d_i - \sum_{j<i} A_{ij} x_j^{(k+1)} - \sum_{j>i} A_{ij} x_j^{(k)}}{A_{ii}}.$$

The first sum uses already updated entries; the second uses entries from the previous sweep. Relaxation gives

$$x_i^{\text{relaxed}} = x_i^{(k)} + \omega \left(y_i^{(k+1)} - x_i^{(k)}\right).$$

Projection then enforces the obstacle:

$$x_i^{\text{new}} = \max\left(\Phi_i, x_i^{\text{relaxed}}\right).$$

The parameter $\omega \in (0, 2)$ controls relaxation. Values above one may accelerate convergence for suitable systems, but excessive relaxation can destabilize the iteration.

2.3.4 Stopping and complementarity diagnostics

The implemented iteration stops when

$$\|x^{(k+1)} - x^{(k)}\|_\infty < \text{tol},$$

or reports failure after the maximum number of sweeps. A small update is necessary but is not the full financial diagnostic. After solving, define

$$z = Ax - d.$$

The LCP checks are

$$x - \Phi \geq 0, \qquad z \geq 0, \qquad (x - \Phi)^{\top} z \approx 0.$$

These separately test obstacle feasibility, residual feasibility, and complementarity.

> **Reader intuition.** PSOR alternates two actions: move toward the linear-system solution, then clip the result back to the financially admissible side of the payoff. Convergence is credible only when both the numerical update and the LCP residual checks are small.

2.4 Implementation Details

The tridiagonal operator is exposed by `src/solvers/cn_psor.py` for sign and matrix tests. Boundary formulas live in `src/solvers/boundary_conditions.py`. The code uses the payoff as the initial condition at $\tau = 0$, forms A and B once per solve, adjusts the first and last right-hand-side entries at every time step, and applies PSOR only when American projection is enabled.

2.5 Validation and Diagnostics

The existing solver audit tests verify the operator on a linear function, the signs in A and B, put and dividend-call boundary values, and PSOR projection/complementarity. European closed-form and no-dividend call controls provide independent checks beyond internal matrix consistency.

2.6 Results

The verified smoke-grid maximum error is 2.782×10^{-4} for both the European put validation and the no-dividend American call control. The American put obstacle violation is reported as zero, with complementarity error 4.009×10^{-12}. These are engineering-scale validation results, not a claim of universal or publication-grade accuracy.

2.7 Failure Modes and Limitations

Uniform grids resolve the payoff kink and free boundary only approximately. Finite-domain truncation can contaminate the interior if $S_{\max}$ is too small. Crank–Nicolson begins from a nonsmooth payoff, and Rannacher smoothing is not active in the current solver. These limitations are particularly important for Gamma near maturity.

2.8 Conclusion and Next Step

The PDE has become a boundary-adjusted LCP at each time step. Later stages extract the exercise boundary and Greeks, study resolution and runtime, and use only diagnostic-passing solver outputs for machine-learning labels.

3 Stage 3: Boundary Extraction and Greeks

3.1 Purpose and Method

Once the CN/PSOR surface is available, Stage 3 derives secondary quantities without pretending that they are as numerically stable as price. For every time level, compute

$$C_i^n = U_i^n - \Phi(S_i).$$

Nodes with C_i^n at or below a small tolerance are classified as exercise nodes; larger premiums indicate continuation.

3.2 Boundary orientation and status

For a put, the exercise set begins at low spot. The algorithm finds the first continuation node and interpolates between it and the preceding exercise node. For a dividend-paying call, exercise is on the high-spot side. The algorithm finds the last continuation node and interpolates toward the next exercise node. These directions are deliberately different.

The output records one of three statuses:

- `boundary_found`: both exercise and continuation nodes exist;
- `no_exercise`: no exercise node appears on the truncated grid;
- `all_exercise`: every grid node is classified as exercise.

If no boundary exists, the code records a missing value rather than fabricating a threshold.

> **Common confusion.** A no-dividend American call control should generally have no early-exercise boundary. Replacing its missing boundary with zero or $S_{\max}$ would turn a valid theorem-based control into a false supervised label.

3.3 Delta, Gamma, and strict masks

For interior nodes, the teaching approximations are

$$\Delta_i^n \approx \frac{U_{i+1}^n - U_{i-1}^n}{2\Delta S},$$

$$\Gamma_i^n \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{(\Delta S)^2}.$$

Delta is a first derivative; Gamma is a second derivative. Differentiation amplifies grid and interpolation error, and the second derivative amplifies it again. The payoff kink at $S = K$ and the exercise transition at $S = B(\tau)$ are therefore delicate regions.

The strict Greek mask excludes nodes near zero spot, the payoff kink, and a finite band around the free boundary. The mask does not make Gamma exact. It defines a less singular region in which comparisons are more meaningful.

> **Reader intuition.** Price can look smooth and accurate while Gamma oscillates, because many nearby price errors cancel in the level but become magnified by repeated differencing.

4 Stage 4: Grid, Domain, and Runtime

4.1 Why resolution matters

A finite-difference answer depends on ΔS, $\Delta\tau$, and $S_{\max}$. Refining the grid reduces discretization error but increases matrix size and the number of time steps. Boundary locations are especially grid-limited because the exercise transition is first detected between neighboring spot nodes.

The grid-240 solution is used as an internal numerical reference. It is not an analytic truth surface. If Q_h is a quantity computed on grid h, the reported comparison has the form

$$\mathrm{RMSE}(Q_h, Q_{\mathrm{ref}}) = \left(\frac{1}{|\Omega|} \sum_{(i,n)\in\Omega} \left(Q^n_{h,i} - Q^n_{\mathrm{ref},i}\right)^2 \right)^{1/2},$$

after interpolation where required. For Greeks, Ω is restricted by the strict mask.

4.2 Domain truncation

The true Black–Scholes spot domain is unbounded, whereas the computation stops at $S_{\max}$. If $S_{\max}$ is too small, the imposed upper boundary can affect economically relevant interior nodes. Comparing several multiples of K tests this sensitivity but does not prove domain independence for every future parameter regime.

4.3 Convergence and runtime interpretation

Price generally converges more smoothly than the free boundary, Delta, or Gamma. Gamma converges most slowly because it is a second derivative and is sensitive to nonsmooth initial data. Runtime rises with the number of time steps, the number of spatial unknowns, and the PSOR sweeps required at each step. The project reports runtime together with iteration counts so that speed is not separated from solver convergence.

> **Common confusion.** A small error relative to grid 240 means agreement with a finer internal calculation. It does not mean agreement with an exact American-option formula, because no such general closed form is being used as the reference.

5 Stages 5–6: Risk Surfaces and Dataset Governance

5.1 What a risk surface represents

A risk surface evaluates price or a diagnostic over many spot, time, and model parameter combinations rather than at one contract point. This reveals how early exercise and sensitivities vary across regimes.

The application sweep separates:

- American puts, whose exercise region is on the low-spot side;
- dividend-paying American calls, which may exercise on the high-spot side;
- no-dividend American call controls, which test the no-early-exercise theorem rather than provide dividend-call boundary evidence.

The verified application set contains 288 regimes: 144 puts, 108 dividend-paying calls, and 36 no-dividend controls. Thirty-two regimes carry diagnostic failures and remain visible in the audit.

5.2 Audit data versus clean labels

Audit retention and model-label eligibility serve different purposes. A failed or delicate regime remains scientifically important because it shows where the solver diagnostics or domain assumptions need attention. It is excluded from the clean machine-learning labels because training on a questionable target would transfer numerical failure into the surrogate.

After controls and flagged regimes are excluded, the clean dataset contains 223 regimes and 3,264,943 samples. The 65 excluded regimes are recorded rather than deleted from the project history.

5.3 Why the split is by regime

All spot-time nodes generated from one parameter tuple share the same option surface. A pointwise random split could place neighboring points from the same surface in both training and test sets. The model could then appear to generalize while mostly interpolating within a regime it has already seen.

A regime-level split assigns the entire surface to one partition. The verified split contains 156 training regimes, 33 validation regimes, and 34 test regimes.

> **Reader intuition.** Treat one parameter regime like one patient in a medical dataset: many measurements from the same underlying case should not be divided between training and testing.

6 Stages 7–8: Direct Price and Positive Premium

6.1 Why low RMSE is insufficient

A direct-price regressor learns

$$\widehat{U}_\theta = f_\theta(x).$$

Standard squared-error training rewards average price accuracy but does not encode $\widehat{U} \geq \Phi$. A model can therefore achieve low RMSE while predicting below the immediate payoff at some nodes.

A soft obstacle penalty may add

$$\lambda \left[\max\Big(\Phi - \widehat{U}, 0\Big)\right]^2$$

to the loss. This discourages violations but remains a trade-off with the data loss. Finite λ, finite optimization, and unseen inputs mean the penalty is not a hard guarantee.

6.2 Positive continuation-premium reconstruction

Instead, predict a nonnegative premium:

$$\widehat{C}_\theta = \text{softplus}(g_\theta(x)) \geq 0,$$

$$\widehat{U} = \Phi + \widehat{C}.$$

Therefore

$$\widehat{U} = \Phi + \widehat{C}, \quad \widehat{C} \geq 0 \Rightarrow \widehat{U} \geq \Phi.$$

This removes one specific failure mode: payoff-obstacle violation.

> **Common confusion.** Obstacle consistency is not full financial correctness. The construction does not enforce monotonicity, convexity, boundary accuracy, Delta accuracy, or Gamma accuracy. It also does not make the neural model a substitute for the CN/PSOR benchmark.

The selected positive-premium MLP has validation RMSE 0.0200, test RMSE 0.0218, and zero obstacle violation on evaluated samples. The linear premium sanity baseline has lower aggregate RMSE in this run, so the neural model is not claimed to dominate all baselines.

7 Stages 9–10: Boundary and Delta Diagnostics

7.1 Why boundary prediction is difficult

The free boundary is an implicit transition derived from an entire option surface. It changes orientation by option type, may be absent, and is limited by the finite-difference grid. Boundary labels are also repeated over many spot rows for the same regime-time pair, so a large point count does not imply an equally large number of independent boundaries.

The boundary model is therefore a separate supervised diagnostic head:

$$\widehat{B}/K = h_\theta(\tau/T, r, q, \sigma, T, \text{option type}).$$

It does not solve the obstacle PDE or reproduce PSOR iterations. The pooled boundary MLP test RMSE is 0.3513 and its test MAE is 0.2087, making it the weakest of the three neural components.

7.2 Why price autograd Delta requires validation

Let

$$y = \frac{U}{K}, \qquad z = \log(S/K).$$

Because $U = Ky$ and

$$\frac{\partial z}{\partial S} = \frac{1}{S},$$

the chain rule gives

$$\Delta = \frac{\partial U}{\partial S} = K\frac{\partial y}{\partial z}\frac{1}{S} = \frac{\partial (U/K)}{\partial \log(S/K)}\frac{1}{S/K}.$$

A differentiable price network supplies this derivative computationally, but price-loss accuracy does not directly train the derivative. Small level errors can still create material slope errors.

The price-model autograd Delta test RMSE is 0.0977, which motivates a separate supervised head. Its output map imposes

$$\Delta_{\text{put}} \in [-1, 0], \qquad \Delta_{\text{call}} \in [0, 1].$$

The bounded Delta MLP has strict-mask test RMSE 0.0332, test MAE 0.0140, and zero bound violation on evaluated samples.

> **Common confusion.** Bounding Delta limits its range; it does not prove that the predicted slope is correct. It also gives no second-derivative control and therefore does not support a Gamma surrogate.

8 Stage 11: Integrated Scenario Screening

8.1 Why the workflow has three heads

The integrated workflow keeps separate mathematical responsibilities:

1. the positive-premium head estimates price while enforcing the payoff obstacle;
2. the boundary head estimates a PSOR-derived transition as a diagnostic;
3. the bounded Delta head estimates a first derivative within option-specific ranges.

Combining their outputs in one interface does not turn them into one universal model. Each head has separate labels, losses, metrics, and limitations.

8.2 Scenario screening, not replacement

Scenario screening means generating rapid preliminary estimates for cases inside the tested regime envelope. Fresh CN/PSOR remains required for new regimes, final label generation, and audit-quality decisions. The integrated experiment evaluates six deterministic held-out regimes: three puts and three dividend-paying calls.

Across those cases, the verified mean price RMSE is 0.0092, mean boundary MAE is 0.1539, and mean strict-mask Delta RMSE is 0.0300. Obstacle and Delta-bound violations are zero on the audited cases. Six cases, all from the designed test envelope, are not enough to establish universal generalization.

Reader intuition. The workflow is a fast screening layer followed by a numerical audit, much as a preliminary model can prioritize scenarios before a higher-fidelity simulation is run.

8.3 Overall Failure Modes and Limitations

The expanded workflow retains the original conservative boundaries:

- grid 240 is an internal reference, not analytic truth;
- Rannacher smoothing is not active;
- boundary estimation remains weaker than price and Delta estimation;
- strict masks exclude the most delicate Greek regions;
- the evidence does not support a Gamma surrogate;
- the neural workflow does not replace CN/PSOR;
- six integrated audit regimes do not establish broad generalization.

8.4 Conclusion and Next Step

The project demonstrates a benchmark-first learning workflow: formulate the obstacle problem, validate CN/PSOR, attach diagnostics, govern labels, then evaluate modular neural approximations under explicit claim limits. Stronger claims would require broader independent regimes, higher-resolution numerical references, and additional derivative validation.